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THEORY EXAMINATION
Question Paper

Roll No. 123109036.

Month and Year of the Examination: May 2025

Semester: 4th

Programme : **B.Tech.**

Subject.....Data Communication & Networking

Course No.....IIPC-212

Number of Questions to be attempted...5

Total No. of Questions.....6

Maximum Marks.....50

Time allowed.....3 HRS

Total No. of Pages...2

Assume any necessary supplementary data, if not provided. Attempt the questions in order. Write answers with help of proper diagrams. Symbols have usual meaning in context of subject.

Ques 1.a	How do the layers of the Internet (TCP/IP) model correlate to the layers of the OSI model?	6
Ques 1.b	For n devices in a network, what is the number of cable links required for a mesh, ring, bus, and star topology?	4
Ques 2.a	What is the maximum data rate of a channel with a bandwidth of 200 KHz if we use four levels of digital signalling?	3
Ques 2.b	A signal with 200 milliwatts power passes through 10 devices, each with an average noise of 2 microwatts. What is the SNR? What is the SNR_{dB} ?	3
Ques 2.c	Compare and contrast a circuit-switched network and a packet-switched network.	4
Ques 3.a	Distinguish between forward error correction versus error correction by retransmission.	5
Ques 3.b	A system uses the Stop-and-Wait ARQ Protocol. If each packet carries 1000 bits of data, how long does it take to send 1 million bits of data if the distance between the sender and receiver is 5000 km and the propagation speed is 2×10^8 m? Ignore transmission, waiting, and processing delays. We assume no data or control frame is lost or damaged.	5
Ques 4.a	In a bus CSMA/CD network with data rate of 10 Mbps, a collision occurs $20 \mu s$ after the first bit of frame leaves the sending station. What should be the length of frame be so that sender can detect collision.	5

Ques 4.b	What are the differences between classful addressing and classless addressing in IPv4? List the classes in classful addressing and define the application of each class (unicast, multicast, broadcast, reserve).	5
Ques 5.a	Compare the TCP header and the UDP header. List the fields in the TCP header that are missing from UDP header. Give the reason for their absence.	5
Ques 5.b	Explain the role and functioning of the Domain Name System (DNS) in the context of internet communication. Describe how a DNS query is resolved, including the key components involved.	5
Ques 6.a	We need a dataword of at least 11 bits. Find the values of k and n in the Hamming code $C(n, k)$ with $d_{\min} = 3$.	5
Ques 6.b	Describe major classes of guided media? List advantages of optical fiber over twisted pair and co-axial cable in tabular format.	5